

Matrix Multiplication

↳ if $A_{p \times q}$ and $B_{m \times n}$, then for AB ,
 q must equal to m . So, $AB_{p \times n}$

↳ If A has n independent columns in C , then $A = CR = (m \times n)(n \times n)$

↳ Matrix multiplication is not commutative. In general, $AB \neq BA$

↳ Matrix associative law is true. $(AB)C = A(BC)$

↳ For rank one matrix:

$$A = C R \quad \begin{array}{l} \rightarrow \text{that one independent row} \\ \rightarrow \text{that one independent column} \end{array}$$

Ex:

$$A = \begin{bmatrix} 1 & 2 & 10 & 100 \\ 3 & 6 & 20 & 200 \\ 2 & 4 & 30 & 300 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 10 & 100 \end{bmatrix}$$

↳ C contains a full set of n independent columns in A .

$R = [I \ F] \rightarrow$ identity matrix I in the same n columns that held C .

↳ The dependent columns of A are combinations of the independent columns in C .

Example 1:

$$A = \begin{bmatrix} 2 & 6 & 4 \\ 4 & 12 & 8 \\ 1 & 3 & 5 \end{bmatrix}$$

Find the independent columns and put them into C

$$C = \begin{bmatrix} 2 & 4 \\ 4 & 8 \\ 1 & 5 \end{bmatrix}, \text{ Then how to find } R?$$

$$\hookrightarrow R = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \left\{ \begin{array}{l} 1. \text{ Match the independent columns with identity matrix column serial} \\ 2. \text{ Notice that the dependent column is 3 times of the first one.} \end{array} \right.$$

$$\hookrightarrow R = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left\{ \begin{array}{l} 1. \text{ Match the independent columns} \\ \text{with identity matrix column serial} \\ 2. \text{ Notice that the dependent column} \\ \text{is 3 times of the first one.} \end{array} \right.$$

Point to be noted that,

1. C contains a full set of π independent columns in A .
2. $R = [I \ F]$ contains the identity matrix I in the same π columns that held C .
3. The dependent columns of A are combinations (CF) of the independent columns in C .

$$\begin{aligned} \text{So, } A &= CR \\ &= C [I \ F] \\ &= [C \ CF] \\ &= \begin{bmatrix} \text{independent columns} & \text{dependent columns} \\ \text{of } A & \text{of } A \end{bmatrix} \end{aligned}$$

To be continued to 3.2